Computer Networks 1
Lab 4

Socket Programming in Java: Chat Application

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I. Objectives

1. Practice with Socket programming in Java
2. Build a simple chat application using client-server model
3. Multithreaded application in Java

II. Content

1. Socket programming in Java

Consider the following code:

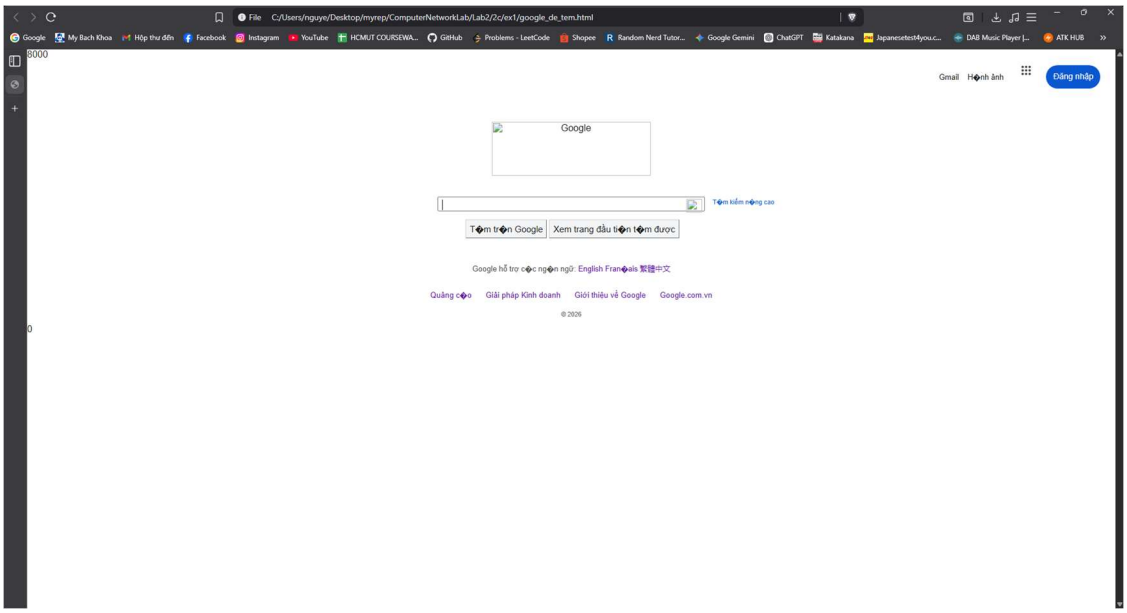
```
try {
      // Get hostname by textual representation of IP address
      InetAddress addr = InetAddress.getByName("127.0.0.1");
// Get hostname by a byte array containing the IP address
byte[] ipAddr = new byte[]{127, 0, 0, 1};   addr =
InetAddress.getByAddress(ipAddr);
      // Get the host name
      String hostname = addr.getHostName();
      // Get canonical host name
      String hostnameCanonical = addr.getCanonicalHostName();
}
catch (UnknownHostException e) {
}
```

Exercise 1:

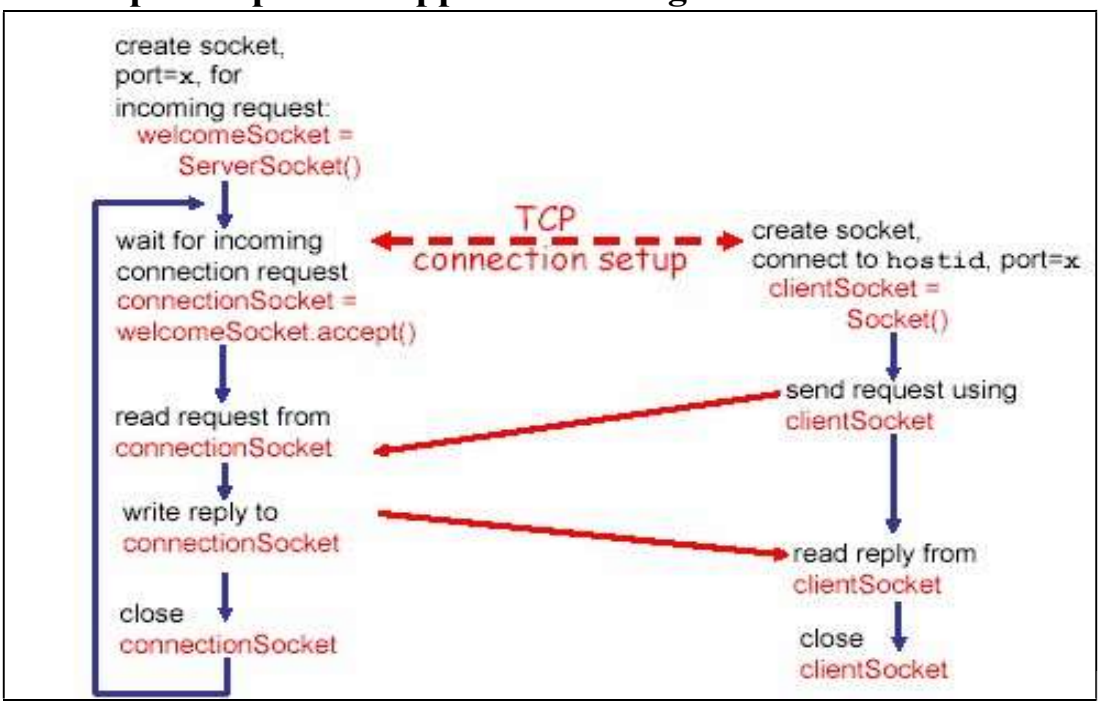
Create a program that connects to a web server and downloads the homepage of this website to local computer.

Ex.: You can test your program with: www.google.com
GitHub Source: 2c/ex1

Demo:



2. Develop a simple chat application using client-server model



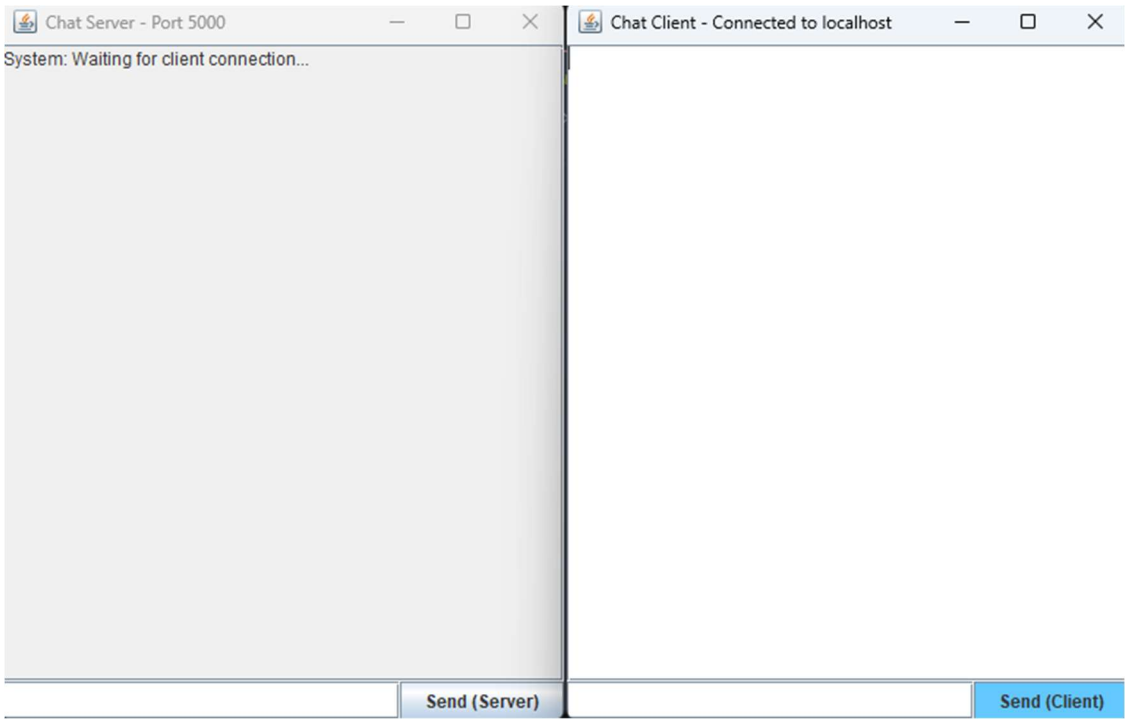
Server application	**Client application**
Create a server socket	
Listen and wait for incoming connection request	Identify server IP and port
	Create a TCP socket to the server
	Establish a connection to the server
Accept the connection request	

Send/Receive data	Send/Receive data
	Close connection
Close connection	

Exercise 2: Design the user interface for the chat application

GitHub Source: 2c/ex2

Demo:



3. Multithread in Java

Create an application that has multiple threads running concurrently:

```
class PrimeRun implements Runnable {
long minPrime;
PrimeRun ( long minPrime ) {
this.minPrime = minPrime;
}
public void run() {
// compute primes larger than minPrime
. . .
}
}
PrimeRun p = new PrimeRun(143);
new Thread(p).start();
```

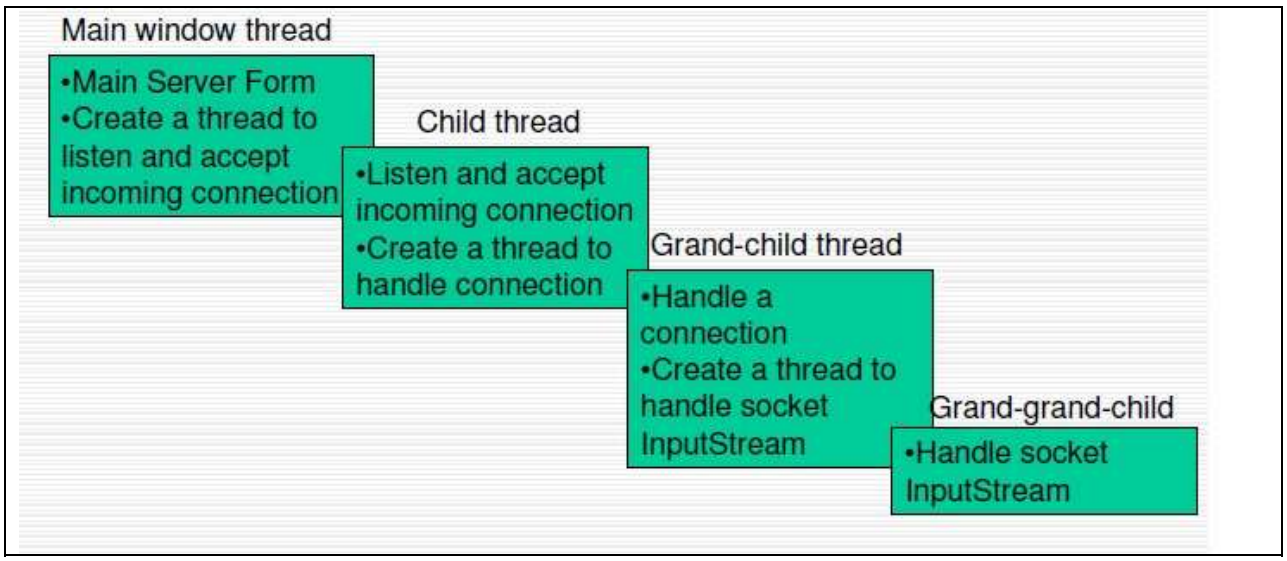
Stop a thread:

Try to use Thread.interrupt() as in the following code and see what happens.

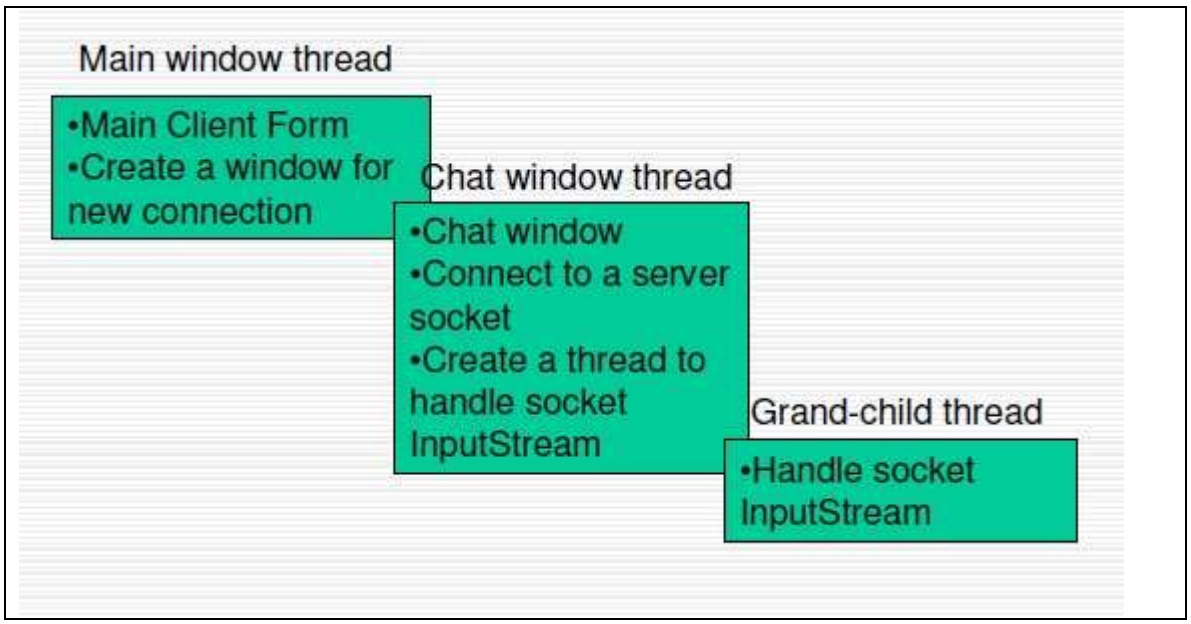
```
public void stop() {
      running = false;
      this.interrupt();
} public void
run() {
      running = true;
      while (running){
            Socket s = socket.accept();
            ...
      }
}
```


```
public void stop() {
      running = false;
      this.interrupt();
} public void
run() {
      running = true;
      while (running){
            Socket s = socket.accept();
            ...
      }
}
```


Multithread in Server Application:

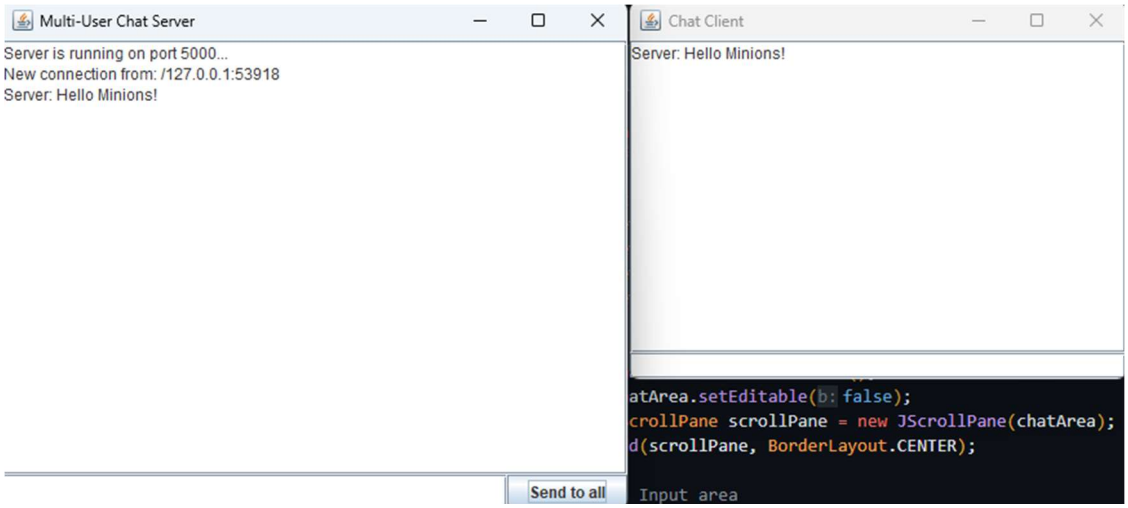


Multithread in Client Application:



Exercise 3:

Using multithread programming model to make the chat application can talk to many different users concurrently.

GitHub Source: [2c/ex3]

Demo: